

Why Sound Change Begins: Pressure, Resistance, and Actuation

Or The Pressure-Resistance-Tipping Mechanism of Linguistic Change Version 2 with examples

This preprint proposes a unified mechanism of sound change based on the interaction of phonetic pressure and sociolinguistic resistance. I argue that changes progress when pressure exceeds resistance, producing a tipping point that triggers rapid adoption and subsequent stabilisation. This pressure–resistance architecture explains why some changes accelerate, why others stall, and why certain shifts reverse direction. I believe this offers a solution for the actuation problem; Actuation occurs at the exact moment when linguistic pressure exceeds resistance, creating a tipping point that allows a change to begin

The model accounts for diverse historical and contemporary phenomena including segment strengthening, weakening, register-specific drift, youth-driven innovation, and identity-policing effects. It replaces fragmented explanations—substrate influence, isolation, prestige, articulatory simplification—with a single causal framework that predicts when and why resistance collapses.

The mechanism is cross-linguistic and generalisable, offering a consistent explanation for changes in Germanic, Celtic, Romance, and other families. By formalising pressure, resistance, tipping, and cascade phases, the model provides a tool for analysing both historical developments and ongoing variation.

This preprint establishes the theoretical foundation for a broader monograph, positioning the mechanism as a candidate for classroom teaching, comparative analysis, and future empirical testing.

Pressure → Resistance → decision point
(Pressure > Resistance?) Tipping → Cascade → Outcome

Case studies

1. English /w/ preservation — did NOT tip

Pressure:

- Labiodental drift (like Dutch /w/ → [ʋ])
- Regional crowding in Kent/Sussex
- Youth articulation pressure
- Contact with varieties using [ʋ]

Resistance:

High functional load of /v/ (Latin/French loans: vine, veil, vast)

Prestige London norm strongly maintained /w/
Minimal pair density made merger costly

Identity policing: labiodental /w/ sounded “rural” or “non-prestige”

Tipping?

No.

Pressure never exceeded resistance because the /v/ contrast was too important.

Cascade:

None — the change stalled.

Outcome:

English kept /w/

Dutch/German tipped;

English resisted

Perfect example of resistance > pressure

2. Spread of /t/ glottalisation in England – did tip

Pressure:

Natural articulatory reduction

Youth identity marking

Urban prestige shift (London → nationwide)

Media amplification (TV, music, influencers)

Resistance:

Older RP norms (“avoid the glottal stop”)

School correction in the 20th century

Association with “working-class speech”

Tipping?

Yes – decisively.

Around late 20th century, pressure > resistance

due to:

Collapse of RP authority

Youth-driven adoption

Media normalisation

Reduced stigma

Cascade

Spread from London → Southeast → Midlands
→ Scotland (although may have originated there
or were both 2 independent changes caused by
the aforementioned articulatory pressures of
word-medial voiceless dental stops)

Entered casual → neutral → formal registers

Became indexical of modern British English

Outcome

Glottalisation is now mainstream

Resistance collapsed

Classic example of pressure tipping → cascade
→ stabilisation

3. Case loss in English — tipped early and cascaded completely

Pressure:

Massive phonological erosion (unstressed vowels → schwa → deletion)

Contact with Norse (simplified morphology)

Loss of inflectional endings

Stress-timing pushing reduction

Resistance:

Morphological system inherited from Old English

Scribal traditions

Prestige of older forms

Tipping?

Yes — very early.

Once endings eroded, resistance had nothing left to protect.

Cascade

Nominative/accusative collapsed

Dative/genitive collapsed

Word order became fixed

Prepositions took over case functions

Entire morphology restructured

Outcome

English became analytic, pressure > resistance

Actuation in more detail:

1. Latent phase:

Pressure builds slowly; resistance still comfortably wins. You see tiny variation, but no directional change.

2. Micro-instability:

Pressure and resistance are closer to balance. You get local fluctuations, youth variants, contact effects—but the system still snaps back.

3. Tipping (actuation):

Net pressure crosses the threshold. The old state is no longer a stable attractor. This is the actuation moment: the first point at which the change is mechanistically forced rather than just possible.

4. Cascade:

Once tipping has happened, diffusion is easy: social propagation, phonological restructuring, rapid spread.

5. Stabilisation:

A new equilibrium forms; resistance now protects the new state.

Why here?

Because this particular speech community has a specific configuration of pressures (contact, phonetics, social dynamics) and resistances (norms, identities, structures) that eventually crosses the threshold.

Why now?

Because pressure is cumulative. It can sit below the threshold for decades, then a small additional push—demographic shift, prestige change, contact event—finally tips the balance.

Why not elsewhere?

Other communities may have similar pressures but stronger resistance, or different population structure, so they never cross the tipping

threshold.

Actuation happens when a pressure–resistance system reaches a critical point where the old state cannot remain stable.

Pressure is common (phonetic biases, social innovation are everywhere).

Resistance is also common (norms, identities, structural fit are everywhere).

Crossing the threshold requires a specific configuration of both.

Societal upheaval weakens resistance, allowing accumulated pressure to cross the tipping threshold; this is why linguistic change accelerates during turbulent periods.

